```
>> three_dof_robot

solve_theta156_left =

[ nx*cos(theta1)*cos(theta6) + ny*cos(theta6)*sin(theta1) - ox*cos(theta1)*sin(theta6) - oy*sin(theta1)*sin(theta6),      ↙
ox*cos(theta1)*cos(theta6) + nx*cos(theta1)*sin(theta6) + oy*cos(theta6)*sin(theta1) + ny*sin(theta1)*sin(theta6),      ↙
ax*cos(theta1) + ay*sin(theta1), px*cos(theta1) + py*sin(theta1) - ay*d6*sin(theta1) - ax*d6*cos(theta1)]
[                                                    nz*cos(theta6) - oz*sin(theta6),   ↙
oz*cos(theta6) + nz*sin(theta6),                    az,                                    pz - d1 - az*d6]
[ nx*cos(theta6)*sin(theta1) - ny*cos(theta1)*cos(theta6) + oy*cos(theta1)*sin(theta6) - ox*sin(theta1)*sin(theta6),      ↙
ox*cos(theta6)*sin(theta1) - ny*cos(theta1)*sin(theta6) - oy*cos(theta1)*cos(theta6) + nx*sin(theta1)*sin(theta6),      ↙
ax*sin(theta1) - ay*cos(theta1), px*sin(theta1) - py*cos(theta1) - ax*d6*sin(theta1) + ay*d6*cos(theta1)]
[                                                                                    0,   ↙
0,                    0,                                                    1]


solve_theta156_right =

[ cos(theta2 + theta3 + theta4)*cos(theta5), -sin(theta2 + theta3 + theta4), -cos(theta2 + theta3 + theta4)*sin(theta5),      ↙
a3*cos(theta2 + theta3) + a2*cos(theta2) + d5*sin(theta2 + theta3 + theta4)]
[ sin(theta2 + theta3 + theta4)*cos(theta5),  cos(theta2 + theta3 + theta4), -sin(theta2 + theta3 + theta4)*sin(theta5),      ↙
a3*sin(theta2 + theta3) + a2*sin(theta2) - d5*cos(theta2 + theta3 + theta4)]
[                    sin(theta5),                    0,                    cos(theta5),   ↙
d4]
[                            0,                    0,                            0,   ↙
1]


solve_theta234_left =

[ nx*cos(theta1)*cos(theta5)*cos(theta6) - ay*sin(theta1)*sin(theta5) - ax*cos(theta1)*sin(theta5) + ny*cos(theta5)*cos      ↙
(theta6)*sin(theta1) - ox*cos(theta1)*cos(theta5)*sin(theta6) - oy*cos(theta5)*sin(theta1)*sin(theta6), ax*cos(theta1)      ↙
*cos(theta5) + ay*cos(theta5)*sin(theta1) + nx*cos(theta1)*cos(theta6)*sin(theta5) + ny*cos(theta6)*sin(theta1)*sin      ↙
(theta5) - ox*cos(theta1)*sin(theta5)*sin(theta6) - oy*sin(theta1)*sin(theta5)*sin(theta6), - ox*cos(theta1)*cos(theta6) -      ↙
nx*cos(theta1)*sin(theta6) - oy*cos(theta6)*sin(theta1) - ny*sin(theta1)*sin(theta6), px*cos(theta1) + py*sin(theta1) -      ↙
ay*d6*sin(theta1) - ax*d6*cos(theta1) + d5*ny*sin(theta1)*sin(theta6) + d5*ox*cos(theta1)*cos(theta6) + d5*nx*cos      ↙
(theta1)*sin(theta6) + d5*oy*cos(theta6)*sin(theta1)]
[                                                                                    nz*cos(theta5)*cos    ↙
(theta6) - az*sin(theta5) - oz*cos(theta5)*sin(theta6),   ↙
az*cos(theta5) + nz*cos(theta6)*sin(theta5) - oz*sin(theta5)*sin(theta6),    ↙
- oz*cos(theta6) - nz*sin(theta6),  ↙
pz - d1 - az*d6 + d5*oz*cos(theta6) + d5*nz*sin(theta6)]
[ ay*cos(theta1)*sin(theta5) - ax*sin(theta1)*sin(theta5) - ny*cos(theta1)*cos(theta5)*cos(theta6) + nx*cos(theta5)*cos      ↙
(theta6)*sin(theta1) + oy*cos(theta1)*cos(theta5)*sin(theta6) - ox*cos(theta5)*sin(theta1)*sin(theta6), ax*cos(theta5)      ↙
*sin(theta1) - ay*cos(theta1)*cos(theta5) - ny*cos(theta1)*cos(theta6)*sin(theta5) + nx*cos(theta6)*sin(theta1)*sin      ↙
(theta5) + oy*cos(theta1)*sin(theta5)*sin(theta6) - ox*sin(theta1)*sin(theta5)*sin(theta6),   oy*cos(theta1)*cos(theta6)      ↙
+ ny*cos(theta1)*sin(theta6) - ox*cos(theta6)*sin(theta1) - nx*sin(theta1)*sin(theta6), px*sin(theta1) - py*cos(theta1) -      ↙
ax*d6*sin(theta1) + ay*d6*cos(theta1) + d5*nx*sin(theta1)*sin(theta6) - d5*oy*cos(theta1)*cos(theta6) - d5*ny*cos      ↙
(theta1)*sin(theta6) + d5*ox*cos(theta6)*sin(theta1)]
[↙
0,↙
0,                                                                                    0,   ↙
1]


solve_theta234_right =

[ cos(theta2 + theta3 + theta4), 0,  sin(theta2 + theta3 + theta4), a3*cos(theta2 + theta3) + a2*cos(theta2)]
[ sin(theta2 + theta3 + theta4), 0, -cos(theta2 + theta3 + theta4), a3*sin(theta2 + theta3) + a2*sin(theta2)]
[                    0, 1,                    0,                    d4]
[                    0, 0,                    0,                    1]

>>
```